

2.7.1 For random variable T in Quiz 2.6, first find the expected value $E[T]$ using Theorem 2.10. Next, find $E[T]$ using Definition 2.14.

Quiz 2.6 Monitor three phone calls and observe whether each one is a voice call or a data call. The random variable N is the number of voice calls. Assume N has PMF

$$P_N(n) = \begin{cases} 0.1 & n = 0, \\ 0.3 & n = 1, 2, 3, \\ 0 & \text{otherwise.} \end{cases} \quad (2.75)$$

Voice calls cost 25 cents each and data calls cost 40 cents each. T cents is the cost of the three telephone calls monitored in the experiment.

(1) Express T as a function of N .

(2) Find $P_T(t)$ and $E[T]$.

2.8.3 Find the variance of the random variable X in Problem 2.4.2.

2.4.2 The random variable X has CDF

$$F_X(x) = \begin{cases} 0 & x < -1, \\ 0.2 & -1 \leq x < 0, \\ 0.7 & 0 \leq x < 1, \\ 1 & x \geq 1. \end{cases}$$

(a) Draw a graph of the CDF.

(b) Write $P_X(x)$, the PMF of X . Be sure to write the value of $P_X(x)$ for all x from $-\infty$ to ∞ .

2.8.5 Let X have the binomial PMF

$$P_X(x) = \binom{4}{x} (1/2)^4.$$

- (a) Find the standard deviation of the random variable X .
- (b) What is $P[\mu_X - \sigma_X \leq X \leq \mu_X + \sigma_X]$, the probability that X is within one standard deviation of the expected value?

2.8.8 Given a random variable X with mean μ_X and variance σ_X^2 , find the mean and variance of the *standardized random variable*

$$Y = \frac{1}{\sigma_X} (X - \mu_X) .$$

3.4.2 Y is an exponential random variable with variance
● $\text{Var}[Y] = 25$.

(a) What is the PDF of Y ?

(b) What is $E[Y^2]$?

(c) What is $P[Y > 5]$?

3.4.4 Y is an Erlang ($n = 2, \lambda = 2$) random variable.



- (a) What is $E[Y]$?
- (b) What is $\text{Var}[Y]$?
- (c) What is $P[0.5 \leq Y < 1.5]$?

3.4.7 The probability density function of random variable X is

$$f_X(x) = \begin{cases} (1/2)e^{-x/2} & x \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is $P[1 \leq X \leq 2]$?
- (b) What is $F_X(x)$, the cumulative distribution function of X ?
- (c) What is $E[X]$, the expected value of X ?
- (d) What is $\text{Var}[X]$, the variance of X ?

- 3.5.1** The peak temperature T , as measured in degrees Fahrenheit, on a July day in New Jersey is the Gaussian $(85, 10)$ random variable. What is $P[T > 100]$, $P[T < 60]$, and $P[70 \leq T \leq 100]$?

- 3.5.4** A function commonly used in communications textbooks for the tail probabilities of Gaussian random variables is the complementary error function, defined as

$$\text{erfc}(z) = \frac{2}{\sqrt{\pi}} \int_z^{\infty} e^{-x^2} dx.$$

Show that

$$Q(z) = \frac{1}{2} \text{erfc} \left(\frac{z}{\sqrt{2}} \right).$$

3.5.8 In this problem, we verify that for $x \geq 0$,

$$\Phi(x) = \frac{1}{2} + \frac{1}{2}\operatorname{erf}\left(\frac{x}{\sqrt{2}}\right).$$

(a) Let Y have a $N(0, 1/2)$ distribution and show that

$$F_Y(y) = \int_{-\infty}^y f_Y(u) \, du = \frac{1}{2} + \operatorname{erf}(y).$$

(b) Observe that $Z = \sqrt{2}Y$ is $N(0, 1)$ and show that

$$\begin{aligned}\Phi(z) &= F_Z(z) = P\left[Y \leq z/\sqrt{2}\right] \\ &= F_Y\left(z/\sqrt{2}\right).\end{aligned}$$